

Verification Environment

Open-Source Based

Technology Stack Overview

- Linux Environment
- Verification Tools
- Source code / Git

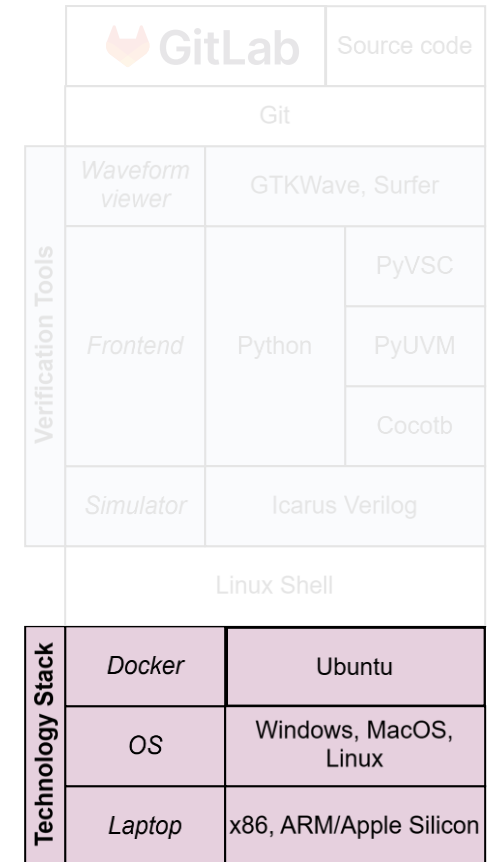
 GitLab		Source code	
Git			
Verification Tools	<i>Waveform viewer</i>	GTKWave, Surfer	
	<i>Frontend</i>	Python	PyVSC
			PyUVM
			Cocotb
	<i>Simulator</i>	Icarus Verilog	
Linux Shell			
Technology Stack	<i>Docker</i>	Ubuntu	
	OS	Windows, MacOS, Linux	
	<i>Laptop</i>	x86, ARM/Apple Silicon	

Linux Environment



Linux Environment

- Toolchain runs in Linux
- In this course: Linux environment in Docker
 - **Docker**: virtualization platform
 - **Dockerfile**: Describes how to build a *Docker image*
 - **Docker Image**: Read-only template for env. with all dependencies
 - **Docker Container**: Isolated instance running an *image*
 - **Docker Volume**: Persistent storage, even if *container* is deleted



Navigating A Linux Shell

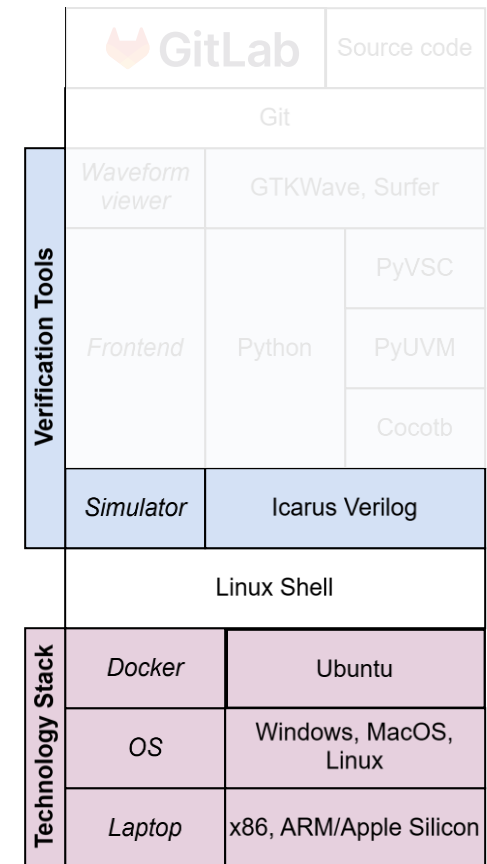
- Docker container has no desktop
 - Some knowledge of shell commands necessary
 - Standard shell in container: Bash
 - Installed editors:
 - Vim
 - Emacs
 - Nano
 - See
 - [verification_env_setup/verification_env_setup.pdf](#)
- Appendix 1 for more info on Terminal basics

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Verification Tools

Simulators

- You may be used to proprietary tools like Xilinx Vivado or Intel Quartus. These perform simulation, synthesis, place-and-route and more in the same program, but also commercial tools like:
 - VCS, Synopsys
 - Questa, Siemens-EDA
 - Xcelium, Cadence
- This is not the general case when using Open-Source
- Must use multiple inter-operating tools to achieve same functionality
- Simulators: The biggest open-source projects are Icarus **Verilog** and **Verilator**



Icarus Verilog

- iverilog tool converts RTL to intermediate representation (IR)
- Not something you will hand-write

```
module dff (  
    input clk, reset,  
    input logic [7:0] d,  
    output logic [7:0] q  
);  
  
always_ff @(posedge clk) begin  
    if (reset) begin  
        q <= '0;  
    end else begin  
        q <= d;  
    end  
end  
  
endmodule: dff
```



```
:ivl_version "11.0 (stable)" "(v11_0-132-gb2f2414f4)";  
:ivl_delay_selection "TYPICAL";  
:vpi_time_precision + 0;  
:vpi_module "/home/jacob/nobackup/icarus/install/lib/ivl/system.vpi";  
:vpi_module "/home/jacob/nobackup/icarus/install/lib/ivl/vhdl_sys.vpi";  
:vpi_module "/home/jacob/nobackup/icarus/install/lib/ivl/vhdl_textio.vpi";  
:vpi_module "/home/jacob/nobackup/icarus/install/lib/ivl/v2005_math.vpi";  
:vpi_module "/home/jacob/nobackup/icarus/install/lib/ivl/va_math.vpi";  
:vpi_module "/home/jacob/nobackup/icarus/install/lib/ivl/v2009.vpi";  
S_0xffae10 .scope package, "$unit" "$unit" 2 1;  
    .timescale 0 0;  
S_0xffafc0 .scope module, "dff" "dff" 3 1;  
    .timescale 0 0;  
        .port_info 0 /INPUT 1 "clk";  
        .port_info 1 /INPUT 1 "reset";  
        .port_info 2 /INPUT 8 "d";  
        .port_info 3 /OUTPUT 8 "q";  
  
... (50 lines total)
```

Icarus Verilog

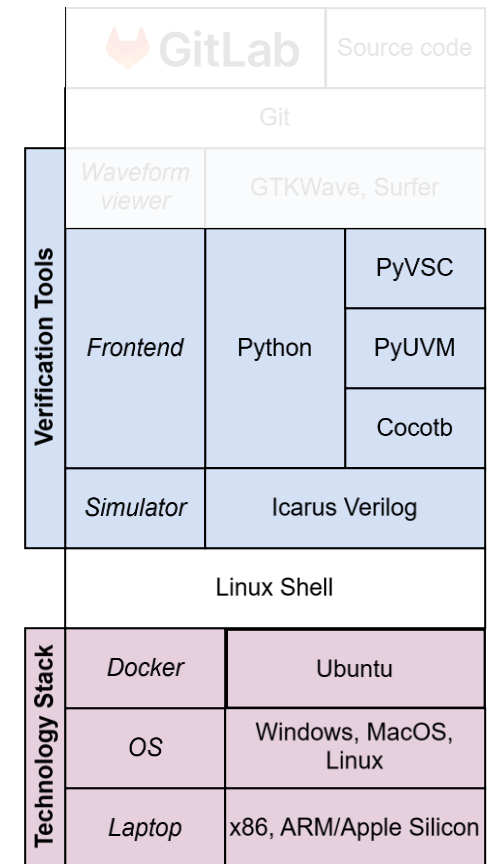
- To convert to VVP format
 - `iverilog -g2012 -o hello.vvp hello.sv`
 - `-g2012`: Use the SystemVerilog 2012 standard
 - `-o hello.vvp`: Set output file to hello.vvp
 - `hello.sv`: Use hello.sv as input file

```
module hello ();  
  initial begin  
    $display("Hello, world!");  
    $finish;  
  end  
endmodule: hello
```

```
$ iverilog -g2012 -o hello.vvp hello.sv #Convert to VVP  
$ vvp hello.vvp #Execute  
Hello, world!
```

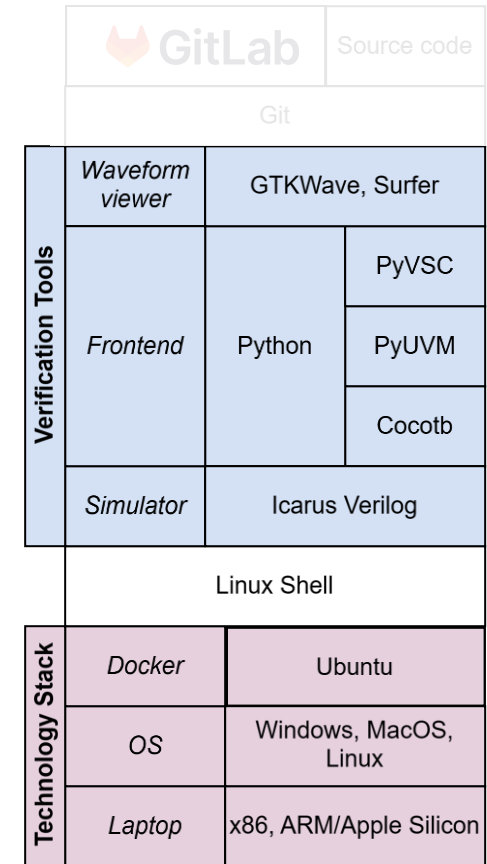
Frontend

- You don't have to worry about the specifics of the simulator that you're using, or how to write wrapper modules
- Cocotb – Interaction with simulator from Python
- PyUVM – Python Implementation of UVM
- PyVSC – Constrained Randomization and Coverage



Waveform Viewers

- Waveforms are essential to debugging errors in hardware designs
- Graphical overview of the signals and their temporal relationships
- Open-source simulators do not have a GUI for viewing waveforms.
- Signal transitions are exported to Value Change Dump (VCD) and viewed with waveform viewer



Waveform viewers

- GTKWave
 - The *de-facto* standard open-source waveform viewer
 - Very powerful, fast and simple to use
- <https://github.com/gtkwave/gtkwave>
- Surfer
 - New, promising tool for viewing waveforms
 - Not as many features, but active open-source development
- <https://gitlab.com/surfer-project/surfer>

GitLab

GitLab

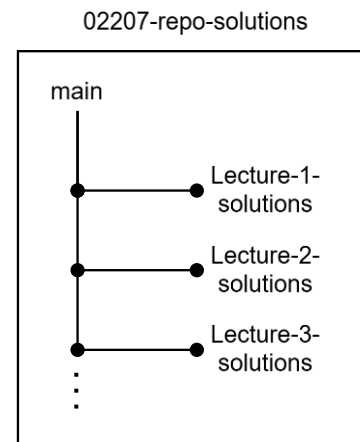
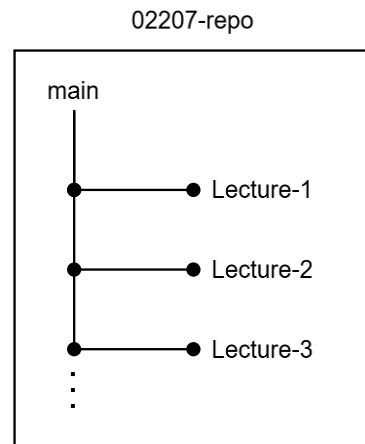
- Code for this course will be shared through DTU GitLab
 - <https://gitlab.gbar.dtu.dk/>
- New code will be released weekly
- Work locally or on your own GitLab repository
- See `verification_env_setup.pdf` Appendix 2 for more info on git

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Gitlab



- Two repositories:
 1. 02207-repo
 2. 02207-repo-solutions
- Each week's state preserved in a dedicated branch



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Linux Environment Setup

Linux setup

- Download `verification_env_setup.zip` from DTU learn
- Follow the steps in `verification_env_setup.pdf`

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